Cube-Sum Permutations: Exact Membership, Multiplicity, and an Exponential Lower Bound

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ such that $i+p_i$ is a perfect cube for every index i. We give a computer-assisted resolution of the existence conjecture recorded as OEIS A006063, as well as the exact membership classification for A096680: $P(n)>0$ for every $n\geq 173$, while $P(173)=P(174)=1$ and $P(n)\geq 3$ for every $n\geq 175$. An explicit seven-block interval involution at $n=173$ and only 1844 one-digit cube roots in 191 alternating-path words certify every remaining base size up to 364. An elementary reflection induction covers all larger sizes; the earlier 109-permutation and 60-seed certificates are retained only for independent cross-checking. Exact Hall-set certificates classify the earlier failures, and alternating-cycle certificates establish multiplicity. For every fixed $s\geq 2$ we prove an s-branch reflection-entropy inequality, giving the stronger limit

$$\liminf_{n\to\infty}\frac{\log P(n)}{(2n)^{1/3}}\geq\frac{\log 2}{2}.$$

A conditional analogue holds for all perfect-power exponents. All finite data and exact checkers accompany this note. Literature priority and external independent review remain open.

1 The reflection principle

Write $\mathcal{C}=\{k^3:k=2,3,\dots\}$ and let $P(n)$ be the number of cube-sum permutations of $\{1,\dots,n\}$. By convention the empty permutation is permitted only in the general extension lemma; our classification concerns $n\geq 1$. Let $\mathcal{G}=\{n\geq 1:P(n)>0\}$.

Lemma 1 (Reflection and counting)**.** *If $0\leq m<n$ and $c=m+n+1$ is a cube, every valid permutation p on $\{1,\dots,m\}$ extends to a valid permutation q on $\{1,\dots,n\}$ via*

$$q_i=\begin{cases}p_i, & i\leq m,\\ c-i, & m<i\leq n.\end{cases}$$

For different admissible cubes c, the images of these extensions are disjoint. In particular,

$$P(n)\geq\sum_{\substack{c\in\mathcal{C}\\ n+1\leq c\leq 2n}}P(c-n-1),\tag{1}$$

with the convention $P(0)=1$.

Proof. Because $c=m+n+1$, reflection $i\mapsto c-i$ permutes precisely the new interval $\{m+1,\dots,n\}$, and $i+q_i=c$ there. Old indices retain their cube sums. The embedding is injective for each c and two different c yield different last entries $q_n=c-n$. The claimed sum counts disjoint families. □

Lemma 2 (The cube-gap bound)**.** *For each $n\geq 365$ some cube c satisfies $n+174\leq c\leq 2n$. For each $n\geq 556$ some cube c satisfies $n+176\leq c\leq 2n$.*

Proof. For $365\leq n\leq 555$, take $c=729$. For $n\geq 556$, let k be the least integer with $k^3\geq n+176$; then $k\geq 10$ and $n\geq(k-1)^3-175$. Thus

$$2n-k^3\geq k^3-6k^2+6k-352\geq 0.$$

The final polynomial equals 108 at $k=10$ and has strictly positive successive differences $3k^2-9k+1$ for $k\geq 10$. Hence the second assertion holds and implies the first for $n\geq 556$. □

2 A short, local existence certificate

The preceding reflection argument is infinite and elementary. The finite existence verification can itself be made much more transparent than an array of 192 long permutations: we give *one* explicit involution at size 173, then 191 short alternating-path words. Only 1844 single-digit cube roots are needed.

Lemma 3 (An explicit order-173 involution)**.** *There is a cube-sum permutation on* $[1, 173]$*, given on each listed interval* I *by* $p(i) = c - i$*. Intervals paired by a double arrow use the same cube* c*:*

intervals I	c
$[1, 7]$	8
$[22, 42]$	64
$[106, 110]$	216
$[8, 12] \longleftrightarrow [15, 19]$	27
$[13, 14] \longleftrightarrow [111, 112]$	125
$[20, 21] \longleftrightarrow [104, 105]$	125
$[43, 103] \longleftrightarrow [113, 173]$	216

Proof. The eleven intervals partition $[1, 173]$. Reflection $i \mapsto c - i$ preserves each of the first three intervals and exchanges the two intervals in each of the last four rows. Thus it defines an involution, and every sum $i + p(i)$ is a cube. □

Lemma 4 (An alternating-word extension)**.** *Suppose* p *is a cube-sum permutation of* $[1, n-1]$*. Let* $w = r_1 \cdots r_t$ *be a finite word of positive integer cube roots. Starting from* $u_0 = n$*, define* $v_j = r_j^3 - u_{j-1}$*. For* $j < t$*, require* $v_j \in [1, n-1]$ *and let* $u_j = p^{-1}(v_j)$*. Suppose all these* $u_0, u_1, \ldots, u_{t-1}$ *are distinct, all* $v_1, \ldots, v_t$ *are distinct, and* $v_t = n$*. Then replacing the old matching edges by*

$$q(u_{j-1}) = v_j \quad (1 \le j \le t),$$

while keeping all other old assignments, yields a cube-sum permutation of $[1, n]$*.*

Proof. The old matching edges removed along this path are $u_j \mapsto v_j$ for $j < t$; these form a bijection between the old visited left and right vertices. Their replacements join u_{j-1} to v_j and have sum r_j^3 by construction. The new left vertex n and the new right vertex n are incorporated, without repeating any endpoint. Every untouched vertex retains its old matching edge. □

The `augmenting_cube_words.txt` appendix lists one word for each $n = 174, \ldots, 364$. Every root is a single decimal digit in $\{2, \ldots, 8\}$. The words are checked using Lemma 4 and the predecessor matching obtained from the previous word; this is *local* verification, not a matching search or a scan to a large cutoff. The independent, standard-library-only `verify_word_paths.py` checks all required edge equalities and endpoint distinctness. The seven-line interval seed plus 1844 root digits certifies all 192 base cases of Theorem 1; combined with the cube-gap induction this proves existence for *every* $n \ge 173$. The older long witness archives remain as secondary independent cross-checks, not as prerequisites of this compact proof.

3 Finite certificates and exact membership

Our finite data are provided as UTF-8 JSON arrays of integers. A valid witness p is checked solely by

$$\operatorname{sort}(p) = (1, \ldots, n), \qquad \forall i \quad i + p_i \in \mathcal{C}.$$

For a negative instance, let $N_n(S) = \{j \in [1, n] : \exists i \in S,\ i + j \in \mathcal{C}\}$. Hall's marriage theorem gives the immediate obstruction

$$|N_n(S)| < |S| \quad \Longrightarrow \quad P(n) = 0. \tag{2}$$

The file `exact_classification_below_173.json` provides 48 explicit valid permutations and 124 explicit Hall-deficient subsets, exactly partitioning $1 \le n \le 172$. Every assertion is directly checked by `verify_extended.py`. For instance, for $n = 172$ the 27 left indices

$$S = \{1, 5, 8, 13, 20, 31, 35, 38, 42, 43, 45, 92, 96, 99,$$
$$104, 111, 122, 126, 129, 133, 134, 136, 153, 157, 160, 165, 172\}$$

have precisely 26 distinct possible right neighbors. Thus $172 \notin \mathcal{G}$, making the threshold below sharp.

For the unbounded tail, only the 109 witness sizes

$$173 \le n \le 255, \qquad 339 \le n \le 364 \tag{3}$$

require explicit stored permutations. Indeed for each $256 \le n \le 338$, put $m = 511 - n \in [173, 255]$; then $m + n + 1 = 512 = 8^3$, so Lemma 1 constructs the missing 83 cases. This compresses the previously saved 192 base cases by $83/192 > 43\%$. All 109 stored witnesses in `core_109.json`, and all 192 reconstructed witnesses, are checked with exact integer arithmetic. The earlier full 192-row file is retained as an additional independent-format audit source.

There is a further useful compression when the 48 verified positive sizes below 173 are reused. An ascending reflection-closure algorithm starts from those small solutions and, at each $173 \le n \le 364$, uses any already certified $m < n$ for which $m+n+1$ is a cube. Precisely 132 lengths are obtained by reflection and only 60 further lengths need explicit witnesses. Here the empty matching is allowed: $n = 342$ follows directly from $c = 343$, with $p(i) = 343 - i$. The 60 additional witnesses are archived in `core_60_with_empty.json`, and `verify_extended.py` certifies this alternate compression independently of the 109-case presentation.

Proposition 1 (Limit of reflection-only compression)**.** *Suppose the empty matching and the* 48 *known positive lengths below* 173 *are provided as initial data. Among proofs of all lengths* $173, \dots, 364$ *that use only the reflection extension of Lemma 1 and independently supplied new full-length seed permutations, at least* 60 *extra seeds are necessary, and* 60 *suffice.*

Proof. Process $n = 173, \dots, 364$ in ascending order. If no cube c yields a previously certified positive predecessor $m = c - n - 1 \in [0, n-1]$, then reflection cannot establish n from earlier data, so a new seed is forced. Otherwise choose such a predecessor and perform reflection, without adding a seed. This deterministic closure rule yields 44 forced seeds in $[173, 255]$, none in $[256, 338]$ (use $c = 512$ and $m = 511 - n$), and 16 in $[339, 364]$. Thus $44 + 16 = 60$, and the attached short integer-only checker reproduces the forced index list. Every forced seed is unavoidable by the definition of the restricted proof rule. □

This is *not* a lower bound on arbitrary mathematical proofs. The alternating-path extension in Lemma 4 is a different operation and indeed replaces those 60 large independent seeds by one explicit involution and a small word table.

Theorem 1 (Complete A006063 membership)**.** *For every integer* $n \ge 173$*,* $P(n) > 0$*. Below* 173*, the complete membership list is*

$$\begin{aligned}\mathcal{G} \cap [1, 172] = \{&7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,\\ &103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121,\\ &122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132,\\ &133, 135, 147, 149, 150, 151, 152, 155, 156, 159, 171\}.\end{aligned}$$

In sequence-index notation this gives

$$\boxed{\text{A006063}(j) = j + 124 \quad (j \ge 49).}$$

Proof. The exact finite witnesses establish $173 \le n \le 364$, and the Hall certificates settle the smaller sizes. For $n \ge 365$, Lemma 2 gives a cube $c \in [n + 174, 2n]$. The predecessor $m = c - n - 1$ satisfies $173 \le m < n$. By strong induction $P(m) > 0$, and Lemma 1 yields $P(n) > 0$. There are precisely 48 positive indices below 173, so the 49th and all later sequence terms are consecutive. □

4 Multiplicity and the second OEIS classification

A cube-sum permutation p is a perfect matching in the bipartite graph with left and right vertices $1, \dots, n$, where (i, j) is an edge exactly when $i + j \in \mathcal{C}$. For any matching p, form a directed graph D_p on the left vertices, putting $i \to j$ whenever $i \ne j$ and (i, p_j) is a cube-sum edge.

Lemma 5 (Alternating-cycle criterion)**.** *For a valid permutation* p*, there exists another distinct valid permutation if and only if* D_p *has a directed cycle.*

Proof. Along a directed cycle $i_1 \to \cdots \to i_t \to i_1$, replace p_{i_s} by $p_{i_{s+1}}$ (cyclic indices). Every moved pair is an allowed edge, and all right vertices remain distinct. Conversely, if $q \neq p$ is another valid permutation, the nontrivial cycles of the permutation $p^{-1} \circ q$ consist of directed cycles in D_p. Thus acyclicity is a rigorous uniqueness certificate. □

The file `second_solutions.json` records directed cycles, not complete duplicate permutations. Direct finite checking establishes: among the 48 positive lengths below 173, exactly

$$E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$$

have a directed alternating cycle; every other positive length below 173, as well as 173 and 174, has an acyclic D_p and therefore a unique permutation.

Furthermore, all 190 base cases $175 \leq n \leq 364$ have *two distinct* alternating-cycle certificates, archived as `second_solutions.json` and `third_solutions.json`. They yield three different permutations, counting the original matching. To extend the multiplicity assertion: for $365 \leq n \leq 553$, the cube 729 gives $m = 728 - n \in [175, 363]$. The two exceptional lengths $554, 555$ have separate pairs of alternating-cycle certificates obtained by extending the unique solutions at $174, 173$ respectively. For $n \geq 556$, the stronger part of Lemma 2 gives $m = c - n - 1 \geq 175$. Three distinct solutions for m remain distinct after reflection, finishing an induction.

Theorem 2 (Complete A096680 membership)**.** *The positive integers with at least two cube-sum permutations are exactly*

$$\boxed{E \ \cup \ \{175, 176, 177, \dots\}.}$$

Moreover $P(n) \geq 3$ for every $n \geq 175$, while $P(173) = P(174) = 1$. The sequence terms satisfy

$$\boxed{\text{A096680}(j) = j + 161 \quad (j \geq 14).}$$

Sharpness of the constant three. The bound $P(n) \geq 3$ for all $n \geq 175$ cannot be improved to four: $P(175) = 3$ exactly. For the stored matching at 175, the directed alternation graph has precisely one nontrivial strongly connected component,

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\},$$

and the other 165 vertices are singleton components. Every cycle of a permutation alternative lies in a strongly connected component, so the permanent reduces to the 10-by-10 component's allowed-edge matrix (including the matching self-edges). A direct bitmask dynamic program gives its permanent as 3. This computation is independently reproduced in `verify_extended.py` using exact integers.

5 A nine-gadget exceptional case at $n = 236$

An *involution* is a permutation satisfying $p^2 = \mathrm{id}$. At $n = 236$ there are exactly 512 cube-sum permutations, *none* of which is an involution. This follows from a short alternating-cycle decomposition, not from a general permanent computation.

Define a reference permutation p of $[1, 236]$ by $p(i) = c - i$ on the following fifteen intervals:

I	c	I	c
$[1, 7]$	8	$[8, 9]$	216
$[10, 18]$	64	$[19, 43]$	125
$[44, 45]$	216	$[46, 70]$	125
$[71, 81]$	216	$[82, 106]$	125
$[107, 134]$	343	$[135, 136]$	216
$[137, 170]$	343	$[171, 172]$	216
$[173, 197]$	343	$[198, 208]$	216
$[209, 236]$	343		

These interval images partition $[1, 236]$, and every prescribed sum is a cube.

Lemma 6 (Translation of alternating cycles)**.** *Suppose $p(i) = c_j - i$ on pairwise disjoint integer intervals I_j for $1 \le j \le r$. If $i_j \in I_j$ form a directed alternating cycle $i_1 \to i_2 \to \cdots \to i_r \to i_1$ for p, then every integer shift t with $i_j + t \in I_j$ for all j produces another alternating cycle on $i_j + t$. If t ranges through a set T of such shifts, the translated cycles are disjoint and yield at least $2^{|T|}$ distinct valid cube-sum permutations.*

Proof. Because $p(i_{j+1} + t) = c_{j+1} - i_{j+1} - t$,

$$(i_j + t) + p(i_{j+1} + t) = i_j + c_{j+1} - i_{j+1} = i_j + p(i_{j+1}),$$

so every translated alternating edge has the same cube sum as its original. Since the I_j are pairwise disjoint, different shifts cannot share a vertex within any I_j. The cycles are therefore vertex-disjoint and can be switched independently. □

In the present 236-element template,

$$10 \to 137 \to 46 \to 198 \to 71 \to 10$$

is one alternating 5-cycle, with successive cube sums

$$216, \quad 216, \quad 64, \quad 343, \quad 125.$$

The five vertices belong respectively to the five different reflection blocks $[10, 18]$, $[137, 170]$, $[46, 70]$, $[198, 208]$, $[71, 81]$. The common shifts $t = 0, \ldots, 8$ produce nine disjoint alternating cycles *by pure algebra*, giving immediately

$$\boxed{P(236) \ge 2^9 = 512}$$

without any matching enumeration. The remaining work for the exact count is only to exclude further cycles.

Lemma 7 (Alternating digraph at 236)**.** *Let D_p be the digraph with an arrow $i \to j$ whenever $i \ne j$ and $i + p(j)$ is a cube. Its strongly connected components comprise* 191 *singletons and exactly nine five-vertex sets*

$$S_t = \{10 + t, 46 + t, 71 + t, 137 + t, 198 + t\}, \qquad 0 \le t \le 8.$$

Within each S_t the nontrivial arrows form one simple directed 5*-cycle, without chords.*

Proof. This is a finite integer check using the only six possible cubes $8, 27, 64, 125, 216, 343$. The interval formula specifies every edge; the independent `count_236.py` verifier checks all 236 rows, the component partition, and the five-cycle properties with integer arithmetic. It invokes neither a matching solver nor a permanent-computation routine. □

Theorem 3 (Exact count and involution obstruction)**.**

$$\boxed{P(236) = 2^9 = 512.}$$

There is no cube-sum involution on $[1, 236]$.

Proof. Given any other cube-sum permutation q, $f = p^{-1} \circ q$ is a permutation whose arrows lie in D_p or are identity loops. Every cycle of f is confined to one strongly connected component. Each singleton is therefore fixed, while each S_t has exactly two choices: five fixed points or the unique directed 5-cycle. The choices are independent, so there are 2^9 permutations q.

For the central set $S_4 = \{14, 50, 75, 141, 202\}$, the reference permutation restricts to

$$14 \mapsto 50 \mapsto 75 \mapsto 141 \mapsto 202 \mapsto 14.$$

The only two possible restrictions of q to S_4 are this 5-cycle and its inverse, neither of which is an involution. Thus no q is an involution. □

The failure of involutivity at 236 is therefore a specific five-cycle obstruction, not a failure of cube-sum permutation existence. The proof uses only a fifteen-row interval template and an easily audited finite graph-decomposition lemma.

6 Many permutations: an exponential cube-root bound

Besides qualitative multiplicity, the reflection family yields a strong quantitative statement. Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$, with the floor defined using integer comparisons (not floating point).

Lemma 8 (Three disjoint embeddings)**.** *If $K(n) = k \geq 27$, each of $k^3, (k-1)^3, (k-2)^3$ gives a reflection extension from a predecessor $m \geq 173$ with $m < n$, and all three predecessors obey $K(m) \geq k - 6$.*

Proof. By definition, $k^3 \leq 2n < (k+1)^3$. For the smallest of these cubes, set $m = (k-2)^3 - n - 1$. Then

$$2m > 2(k-2)^3 - (k+1)^3 - 2 = k^3 - 15k^2 + 21k - 19.$$

The right side exceeds $(k-6)^3$ for $k \geq 27$, since the difference is

$$3k^2 - 87k + 197 > 0 \quad (k \geq 27).$$

Thus $2m > (k-6)^3 \geq 21^3$, which implies $m \geq 173$ and $K(m) \geq k - 6$. Also $(k-2)^3 < k^3 \leq 2n$, so $m < n$. The other two cubes give larger predecessors and satisfy the same bounds. Distinct cubes yield disjoint matching families by Lemma 1. □

Theorem 4 (A096904 quantitative bound)**.** *For all integers $n \geq 173$, with $q(n) = \max\{0, \lfloor (K(n) - 27)/6 \rfloor\}$,*

$$\boxed{P(n) \geq 3^{q(n)}}.$$

In particular

$$\log P(n) \;\geq\; \frac{\log 3}{6}(2n)^{1/3} - O(1),$$

so $P(n)$ grows at least exponentially in $n^{1/3}$.

Proof. If $K(n) < 27$ then $q(n) = 0$, and Theorem 1 supplies $P(n) \geq 1$. For $K(n) \geq 27$, Lemma 8 provides three distinct extensions from smaller predecessors $m_i \geq 173$ with $K(m_i) \geq K(n) - 6$. Thus $q(m_i) \geq q(n) - 1$. By strong induction and the disjointness of the three extension images,

$$P(n) \geq \sum_{i=1}^{3} P(m_i) \geq 3 \cdot 3^{\max(0, q(n)-1)} \geq 3^{q(n)}.$$

The displayed logarithmic asymptotic follows immediately. □

7 A universal reflection-entropy bound

The fixed three-branch construction above is only one instance of a general branching mechanism. In fact, one may use *any fixed number* of the largest available cubes and obtain a sharp limiting coefficient for this family of lower bounds.

For each integer $s \geq 2$, let $\rho_s \in (1, 2)$ be the unique solution of

$$\sum_{h=1}^{s} \rho_s^{-h} = 1, \qquad B_s = 2s^2 + 4s - 1. \tag{4}$$

Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ as before.

Theorem 5 (Reflection entropy)**.** *For all $s \geq 2$ and $n \geq 173$,*

$$\boxed{P(n) \geq \rho_s^{(K(n) - B_s)/2}.} \tag{5}$$

In particular,

$$\boxed{\liminf_{n \to \infty} \frac{\log P(n)}{(2n)^{1/3}} \;\geq\; \frac{\log 2}{2}.} \tag{6}$$

Equivalently, for every $\varepsilon > 0$ there is a constant C_ε such that

$$P(n) \geq \exp\left(\left(\tfrac{1}{2} \log 2 - \varepsilon \right) (2n)^{1/3} - C_\varepsilon \right) \quad (n \geq 173).$$

Proof. Fix s and write $B = B_s$, $\rho = \rho_s$. We prove (5) by strong induction. For $K(n) < B$, the right side is at most 1, and $P(n) \geq 1$ by Theorem 1.

Now suppose $k = K(n) \geq B$. For each $j = 0, \ldots, s-1$, take the cube $c_j = (k-j)^3$ and put $m_j = c_j - n - 1$. Since $k^3 \leq 2n < (k+1)^3$, the following exact polynomial inequality holds:

$$\begin{aligned} 2m_j - (k-2j-2)^3 &\geq 2(k-j)^3 - (k+1)^3 - 1 - (k-2j-2)^3 \\ &= 3k^2 - (6j^2+24j+15)k \\ &\quad + 6j^3 + 24j^2 + 24j + 6 \\ &\geq 0. \end{aligned}$$

Dividing the final polynomial by 3, its coefficient of $-k$ is $2j^2+8j+5$ and its constant term is positive. Hence the last inequality follows from $k \geq B \geq 2j^2+8j+5$. Also $k-2j-2 \geq k-2s \geq 8$, so $m_j \geq 256 \geq 173$ and $K(m_j) \geq k-2(j+1)$. Each $c_j \leq k^3 \leq 2n$, so $m_j < n$.

The s reflection extensions are pairwise disjoint by their distinct last images, as in Lemma 1. Using the strong induction hypothesis, we obtain

$$\begin{aligned} P(n) &\geq \sum_{j=0}^{s-1} P(m_j) \\ &\geq \sum_{j=0}^{s-1} \rho^{(K(m_j)-B)/2} \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

This proves the first assertion. The ρ_s are strictly increasing and converge to 2, since their defining finite geometric sums approach $\sum_{h\geq 1} 2^{-h} = 1$. For each fixed s, B_s is a constant independent of n and $K(n) = (2n)^{1/3} + O(1)$. Therefore the ratio on the left side of (6) is at least $\frac{1}{2}\log\rho_s$ in the limit. Letting $s \to \infty$ proves (6). □

Corollary 1 (Other power sums, conditional on existence)**.** *For an integer $d \geq 2$, let $P_d(n)$ count permutations of $[1,n]$ in which every $i + p_i$ is a perfect dth power. If $P_d(n) > 0$ for all sufficiently large n, then*

$$\liminf_{n\to\infty} \frac{\log P_d(n)}{(2n)^{1/d}} \geq \frac{\log 2}{2}.$$

Proof. For any fixed s and $j = 0, \ldots, s-1$, write $k = \lfloor (2n)^{1/d} \rfloor$ and use $c_j = (k-j)^d$, $m_j = c_j - n - 1$. The binomial theorem gives, uniformly over these finitely many j,

$$2(k-j)^d - (k+1)^d - (k-2j-2)^d = dk^{d-1} + O_{d,s}(k^{d-2}) > 1$$

for all sufficiently large k. It follows that m_j is a smaller positive length in the eventual-existence range and $\lfloor (2m_j)^{1/d} \rfloor \geq k-2j-2$. The distinct reflection families obey the same s-branch induction as Theorem 5, with a suitable constant initial offset in k. The ρ_s tend to 2 as before. □

Theorem 6 (A near-critical lower bound with explicit loss)**.** *For $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ and every $n \geq 173$,*

$$\boxed{P(n) \geq \frac{127}{K(n)}\, 2^{(K(n)-127)/2}.} \tag{7}$$

More generally, for every real $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$\boxed{P(n) \geq C_\varepsilon \frac{2^{K(n)/2}}{K(n)^\varepsilon} \qquad (n \geq 173).} \tag{8}$$

Proof. Write $k = K(n)$. The exact cubic polynomial inequality in the proof of Theorem 5 works whenever

$$k \geq B_s = 2s^2 + 4s - 1, \qquad s \geq 2;$$

it supplies s *disjoint* reflection embeddings with predecessor indices $m_h \in [173, n-1]$ and $K(m_h) \geq k - 2h$ for $h = 1, \ldots, s$.

First set $F(k) = 2^{k/2}/k$, which is increasing for all integer $k \geq 3$. For $k \geq 128$ choose $s = \lfloor \log_2 k \rfloor$. Then $s \geq 7$ and $k \geq 2^s \geq B_s$: the last inequality holds at $s = 7$, since $128 \geq 125$, and is preserved upon doubling, because $2B_s \geq B_{s+1}$ for all $s \geq 7$. Moreover $k - 2s \geq 8$, so every predecessor is in the increasing range of F. By the reflection counting lemma,

$$\begin{aligned}\frac{\sum_{h=1}^{s} F(k-2h)}{F(k)} &= \sum_{h=1}^{s} \frac{2^{-h}k}{k-2h} \\ &\geq \sum_{h=1}^{s} 2^{-h}\left(1 + \frac{2h}{k}\right) \\ &\geq 1 - 2^{-s} + \frac{2}{k} > 1,\end{aligned}$$

using $\sum_{h=1}^{s} h2^{-h} \geq 1$ for $s \geq 2$ and $k < 2^{s+1}$. For $k < 128$, Theorem 1 ensures $P(n) \geq 1$. Because F is increasing for $k \geq 7$, this implies $P(n) \geq F(k)/F(127)$. Strong induction using the above inequality extends the same bound to all $k \geq 128$. Since $F(127) = 2^{127/2}/127$, this is exactly (7).

For the more general result, fix $\varepsilon > 0$ and define $F_\varepsilon(k) = 2^{k/2}/k^\varepsilon$. It is increasing for all sufficiently large k. For sufficiently large k, set $s = \lceil \log_2(k/\varepsilon) \rceil$, so $s = O_\varepsilon(\log k)$, $k \geq B_s$ and $2^{-s} \leq \varepsilon/k$. Each predecessor is in the increasing range of F_ε. By the elementary Bernoulli inequality,

$$\left(\frac{k}{k-2h}\right)^{\varepsilon} \geq 1 + \frac{2\varepsilon h}{k}.$$

Consequently

$$\begin{aligned}\frac{\sum_{h=1}^{s} F_\varepsilon(k-2h)}{F_\varepsilon(k)} &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k}\sum_{h=1}^{s} h2^{-h} \\ &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k} > 1.\end{aligned}$$

Choose an initial K_ε large enough for all these inequalities to hold at $k \geq K_\varepsilon$. By positivity, the bound $P(n) \geq C_\varepsilon F_\varepsilon(k)$ holds for $k < K_\varepsilon$ on taking $C_\varepsilon = 2^{-K_\varepsilon/2}$. The same strong induction proves it for every larger k, yielding (8). □

These bounds approach the *critical* reflection coefficient $\frac{1}{2}\log 2$ with a logarithmic (indeed arbitrarily small polynomial) loss. Their proofs use no new computer-assisted certificate beyond the existence theorem.

The coefficient $(\log 2)/2$ is a *proved lower bound*, not an asserted exact exponential growth rate. The argument needs only the qualitative existence Theorem 1; it makes no further use of numerical matching data.

8 An efficient constructive algorithm

The proof is also algorithmic. Given $n > 364$, choose the *least* cube $c = k^3 \geq n + 174$ and put $m = c - n - 1$. The cube-gap lemma ensures $173 \leq m < n$. Minimality gives

$$(k-1)^3 < n + 174 \leq k^3 \quad \Longrightarrow \quad 173 \leq m \leq 3k^2 - 3k + 173 = O(n^{2/3}).$$

Repeat until the remaining size is in the certified interval $173 \leq m \leq 364$, then fill each recorded interval by the explicit reflection $p_i = c - i$.

Proposition 2 (Linear-output construction)**.** *A valid cube-sum permutation of any $n \geq 173$ can be constructed in $O(n)$ unit-cost arithmetic operations, with $O(\log\log n)$ reflection layers.*

Proof. The predecessor bound $m = O(n^{2/3})$ implies $\log m \leq (2/3)\log n + O(1)$, so the number of iterations before the fixed-size base is $O(\log\log n)$. The returned intervals are disjoint and their union, together with the base interval, is $\{1, \ldots, n\}$. Therefore exactly n entries are written. Each exact integer cube root can be computed using $O(\log n)$ comparisons, which contributes $O(\log n \log\log n) = o(n)$ additional unit-cost operations. □

The accompanying `construct_linear.py` implements this procedure with exact integer cube roots, and can also display the recursion plan without allocating the full output.

9 Reproduction and limitations

All programs require only Python 3 standard-library modules:

```
python papers/A006063/verify.py
python papers/A006063/independent_match.py
python papers/A006063/verify_extended.py
python papers/A006063/construct_linear.py
python papers/A006063/verify_word_paths.py
python papers/A006063/count_236.py
```

The archived data consist of 109 core witnesses, the legacy 192-row all-base cross-check, 48 small positive permutations, 124 negative Hall sets, and compact pairs of alternating-cycle certificates. The infinite inductive and counting lemmas are mathematical arguments rather than extrapolations from numerical experiments.

The status is a *computer-assisted proof candidate* pending independent scrutiny of the finite certificate, its interpretation, and literature priority. The present manuscript does not compute the exact values of $P(n)$ in general, nor does it claim external peer review.

A All 191 short alternating-path words

The integer at the left is the new size n; the following one-digit symbols are the successive cube roots r_j in Lemma 4. Together with the explicit seed in Lemma 3, these words form an independently and locally checkable existence certificate for $173 \le n \le 364$.

174: 6355553636355555363633555536
175: 6336
176: 65555556
177: 6364636
178: 6364625555556
179: 74635555555547
180: 74474247447
181: 6554746474556
182: 655556
183: 663647
184: 66
185: 66264746366
186: 66266
187: 7447
188: 7455665547
189: 666363647447447
190: 647
191: 6665547556
192: 75257
193: 7575555757
194: 66655666
195: 6264757
196: 663666626
197: 75746264757
198: 65575264757
199: 666
200: 66633666
201: 64747574757
202: 6474752556
203: 663466666
204: 757424757
205: 65755756
206: 6666446666
207: 666665546666
208: 64757
209: 6443756
210: 757464757
211: 6666357
212: 66664666
213: 757475757
214: 653756
215: 6666556
216: 7555666666665557
217: 75757375757375757
218: 7555757573573757575557
219: 757455757464757
220: 75757554665575757
221: 757573753666375757
222: 7375757
223: 757455755557554757
224: 7363753637
225: 7557357447557
226: 75447557
227: 757537525735757
228: 7524665566666657
229: 746664667
230: 752757573637557
231: 75666637575725257
232: 757536357556666665757
233: 75666666557537
234: 7375757256666667
235: 77
236: 7466466466466466466475727574 7
237: 7375747
238: 74664747
239: 73444667
240: 7466466444667
241: 7337
242: 734664667
243: 73527537534667
244: 73667
245: 746666667
246: 766375375375266667
247: 7666666666666666667
248: 76666266667
249: 766662757475757266667
250: 747
251: 766665736666667
252: 747573375747
253: 7474466375747
254: 74744664664747
255: 74757377
256: 8
257: 76666667
258: 747525747
259: 85666658566277
260: 85662666666667
261: 7475375758
262: 8558
263: 7736666444758
264: 7737574744758
265: 8558534758
266: 766664758
267: 85585757377
268: 7477
269: 766666677477
270: 8575247477
271: 8575266667
272: 7475375375758575758
273: 855857573757447
274: 85266258
275: 7437447
276: 74586666666658
277: 868
278: 7447
279: 868525377
280: 773368
281: 76666477
282: 72755857477
283: 727
284: 777558
285: 727575868
286: 85475758
287: 76658
288: 7577666658
289: 8666857537
290: 8558
291: 856666667
292: 86468
293: 8558667
294: 737575727737
295: 86477737
296: 7775486477
297: 83747775868
298: 7577738
299: 7775468
300: 8646368585486868
301: 7368666468
302: 76644867
303: 734775557
304: 736868
305: 8677777
306: 7586666856768
307: 7684777558
308: 768458667
309: 77776667
310: 76848
311: 77
312: 836366625868
313: 775868577
314: 77457777
315: 7764777
316: 867738
317: 76664468
318: 7668667
319: 864866367
320: 764777
321: 8685485677
322: 7684847777
323: 75737668
324: 7667677367
325: 7586268
326: 7586847
327: 8685758268
328: 8626846668
329: 7577
330: 86856768667
331: 7585268368
332: 867
333: 777777
334: 7686855758
335: 76358668668
336: 784686847586868
337: 8638486777
338: 7468487
339: 77
340: 77583867
341: 7777577
342: 88288
343: 86388
344: 86867777778
345: 858557288
346: 85777488
347: 8586677768
348: 8855676668
349: 8848
350: 8485738686388
351: 86864485882778
352: 8686488
353: 8577664648
354: 85868684848
355: 877388
356: 84577488
357: 86777755778
358: 858557785588
359: 8468
360: 8586482775858
361: 8586855588
362: 883477778
363: 8528686488
364: 855557388

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